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A Multicenter Clinical Study on Treating Post-Dural Puncture Headache with an Intravenous Injection of Aminophylline

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ABSTRACT / RESUMEN CLï¿½NICO

Background:

Post-dural puncture headache (PDPH) is a common adverse effect following lumbar puncture or epidural breach. Traditional treatments include bed rest, hydration, and epidural blood patch (EBP). This multicenter prospective trial evaluated intravenous aminophylline as a safe and effective therapeutic option.

Study Design:

Prospective multicenter clinical study involving 128 consecutive patients diagnosed with moderate-to-severe PDPH following diagnostic or therapeutic lumbar puncture across 4 tertiary hospitals.

Intervention:

Patients received a single intravenous infusion of 250 mg aminophylline in 100 mL of 0.9% normal saline administered over 30 minutes. Visual Analog Scale (VAS) pain scores were recorded at baseline (0 min), 30 min, 2 h, 24 h, and 48 h.

Results:

The mean baseline VAS score was 7.72 ï¿½ 1.15. Following aminophylline administration, mean VAS scores decreased significantly to 4.84 ï¿½ 1.42 at 30 minutes, 2.31 ï¿½ 1.18 at 2 hours, and 0.81 ï¿½ 0.95 at 48 hours ($P < 0.001$ at all time points). The overall treatment response rate at 48 hours was 93.8% (120/128 patients).

Conclusion:

Intravenous injection of aminophylline provides rapid, dramatic, and sustained pain relief in PDPH with high clinical efficacy and an excellent safety profile, representing an effective first-line therapy before considering an epidural blood patch.

TABLE: PAIN SCORES (VAS) OVER TIME FOLLOWING AMINOPHYLLINE INFUSION

Time Point	Mean VAS Score (ï¿½ SD)	Pain Reduction (%)	P-value vs Baseline
Baseline (0 min)	**7.72 ï¿½ 1.15**	**0%**	**Reference**
30 Minutes Post-Infusion	**4.84 ï¿½ 1.42**	**37.3%**	**< 0.001 *** **
2 Hours Post-Infusion	**2.31 ï¿½ 1.18**	**70.1%**	**< 0.001 *** **
24 Hours Post-Infusion	**1.12 ï¿½ 0.88**	**85.5%**	**< 0.001 *** **
48 Hours Post-Infusion	**0.81 ï¿½ 0.95**	**89.5%**	**< 0.001 *** **

DISCUSSION & CLINICAL SUMMARY

This multicenter study demonstrates that IV aminophylline (250 mg over 30 min) is highly effective for post-dural puncture headache.

Highlights of the study:
1. Speed of Relief: Over 37% reduction in pain intensity within 30 minutes of infusion start, providing rapid symptom control in the acute setting.
2. Sustained Efficacy: At 48 hours, mean VAS score dropped to 0.81, with 93.8% of patients achieving complete or near-complete resolution.
3. Safety: Minimal side effects (mild nausea in 3 patients, mild insomnia in 2 patients) which resolved spontaneously.

Conclusion: Intravenous aminophylline is a safe, simple, economical, and fast-acting treatment for PDPH, making it an optimal first-line agent before attempting invasive epidural blood patching.

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